Research on water consumption for 1 kg beef production

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Water Consumption in Beef Production: A Comprehensive Analysis

Key Points

- Water consumption for 1 kg of beef varies widely, from 20 to 200,000 liters globally, due to calculation methods, production systems, and local factors.
- In Canada, about 15,944 liters of water are estimated for 1 kg of beef, and there was a 17% reduction in water use per unit of Canadian beef in 2011 compared to 1981.
- Over 90% of the water footprint for beef production is green water, while blue and gray water account for 158 gallons per pound.
- Feedlot systems can use up to 552.5 liters of water per kg of hot carcass weight, while semi - intensive silvopastoral systems use as little as 87.2 liters.
- Advanced irrigation technologies like subsurface drip irrigation can reduce irrigation water use by 45% in beef production.

Overview

Water consumption in beef production is a critical issue in the context of global water scarcity and sustainable agriculture. The amount of water required to produce 1 kg of beef is influenced by numerous factors, including production methods, regional differences, and climate change. Understanding these aspects is essential for developing strategies to improve water - use efficiency in the beef industry and minimize its environmental impact.

Detailed Analysis

Global and Regional Variation in Water Consumption

The estimated water used to produce 1 kg of beef shows a vast global range, from 20 to 200,000 liters. Different calculation methods, such as the water footprint and consumptive water footprint, contribute to this wide variation. For example, the water footprint method often estimates around 15,000 liters per kg of beef, while the consumptive water footprint method estimates between 302 and 1337 liters per kg of beef in France.

`![The graphic provides a comparative overview of the gallons of water required to produce one pound of various foods](https://www.denverwater.org/sites/default/files/tap_featured_images/WaterFootprintChart-1.jpg)`

Country - specific examples also illustrate significant regional differences. In Japan, it is about 11,000 liters per kg, in Mexico up to 37,800 liters, and in the United States, about 2479 liters for 1 kg of cooked and consumed beef. The water required for feed, which varies based on geographical location and farming systems, also plays a crucial role. Using water - intensive crops like soybeans increases the water footprint, while legumes like faba beans can reduce it.

Water Consumption in Different Production Systems

Conserved water use varies significantly across beef production systems. Feedlot systems can have a high water usage per kg of hot carcass weight (up to 552.5 liters), while semi - intensive silvopastoral systems use as little as 87.2 liters, and extensive unmodified pasture uses 154.5 liters. These values are calculated based on water drunk by animals, evaporation from troughs, water for irrigation of crops and pasture, and water for fertilizer production.

`![The graphic compares the water usage required to produce one pound of various meats](https://s3.amazonaws.com/tr-web-v2/images/09/61/70a26e05-ef69-4176-8e11-b28f3aa3e5ae/7040000961_d.jpg)`

Types of Water in Beef Production

Over 90% of the water footprint for beef production is green water (rainwater). The blue and gray water footprint is 158 gallons per pound. Blue water is mainly used for crop irrigation and animal drinking, and gray water is for cleaning facilities and processing plants.

`![The table presents the average green and blue water use per kilogram for various animal products](https://meatthefacts.eu/wp-content/uploads/2019/09/environment_5.jpg)`

Water - Efficiency Metrics

Advances in irrigation technology can significantly improve water - use efficiency in beef production. Subsurface drip irrigation can reduce irrigation water use by 45%, and variable rate irrigation adjusts water application based on soil and plant needs.

Future Projections and Climate Change Impact

Global water demand is expected to increase by 50% between 1995 and 2025, which may expand water - scarce areas and impact beef production. Climate change is altering rainfall patterns, which can affect the availability of water for beef production, especially for rain - fed pastures and feed crops.

Stakeholder Perspectives

Although direct stakeholder perspectives were not found, farmers may be interested in water - efficient systems to reduce costs, meat processors may focus on optimizing blue and gray water use, and environmental groups could advocate for sustainable farming practices relying more on green water.

Survey Note

Literature Review & Theoretical Framework

Existing research on water consumption in beef production has primarily focused on quantifying the water footprint using different methods and analyzing the factors influencing water use, such as production systems, feed sources, and regional differences. Theoretical frameworks related to sustainable agriculture and water management have been used to understand the environmental impact of beef production and develop strategies for improvement.

Methodology & Data Analysis

Data on water consumption in beef production are collected from various sources, including agricultural research institutions, industry reports, and academic papers. Different calculation methods are employed, such as the water footprint and consumptive water footprint methods, to estimate water use. Statistical analysis is used to compare water consumption across regions, production systems, and over time.

Critical Discussion

The wide range of water consumption estimates highlights the complexity of measuring water use in beef production. The dominance of green water in the water footprint suggests that sustainable farming practices relying on rainwater can be an effective way to reduce the environmental impact of beef production. However, the importance of blue water for irrigation and the role of gray water in processing facilities cannot be ignored. Future research should focus on improving the accuracy of water - consumption estimates and developing more effective water - efficiency metrics.

Future Research Directions

Future research could explore the potential of emerging technologies, such as precision agriculture and genetic improvement of cattle, to reduce water consumption in beef production. Additionally, more in - depth studies on stakeholder perspectives and their willingness to adopt water - efficient practices are needed. The impact of climate change on water availability and beef - production efficiency also requires further investigation.

Key Citations

- [Successes and Problems with Measuring Water Consumption in Beef Systems ...](https://academic.oup.com/edited-volume/34660/chapter/295330694)
- ["The water footprint" within the cattle industry - nutriNews](https://nutrinews.com/en/the-water-footprint-within-the-cattle-industry/)
- [How Much Water is Used to Make a Pound of Beef? BeefResearch.ca](https://www.beefresearch.ca/fr/blog/cattle-feed-water-use/)
- [It takes 15000 liters of water to produce 1 kg of beef: TRUE or FALSE?](https://chaire-bea.vetagro-sup.fr/en/it-takes-15000-liters-of-water-to-produce-1-kg-of-beef-true-or-false/)

• [A comprehensive environmental assessment of beef production and ...](https://www.sciencedirect.com/science/article/pii/S0959652623009241)

• [Land and Water Usage in Beef Production Systems - PMC](https://pmc.ncbi.nlm.nih.gov/articles/PMC6616661/)

• [Does beef production really use that much water?](https://ksubci.org/2020/11/16/does-beef-production-really-use-that-much-water/)